Quadratics: Expanding & Factorising

Practice test 1 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify (x + 4)(x + 7)

[2 marks]

A2. Expand and simplify (x – 5)(x + 3)

[2 marks]

A3. Expand and simplify $(x – 6)^2$

[2 marks]

A4. Expand and simplify (3x + 2)(2x – 5)

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify (x + 1)(x + 2)(x + 3)

[3 marks]

B2. Expand and simplify $(x – 2)(x + 4)(2x + 1)$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 + 9x + 20$

[2 marks]

C2. Factorise $x^2 – 3x – 28$

[2 marks]

C3. Factorise $x^2 – 49$

[1 mark]

C4. Factorise fully $2x^2 – 50$

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $2x^2 + 7x + 3$

[2 marks]

D2. Factorise $3x^2 – 10x – 8$

[2 marks]

D3. Factorise fully $4x^2 + 10x - 6$

[2 marks]

Section E — Using factorising

E1. Solve $x^2 - 5x - 14 = 0$

[3 marks]

E2. Simplify fully

$$\frac{x^2 + 5x + 6}{x^2 - 9}$$

[3 marks]

Section F — Problem solving & proof

F1. Show that $(n + 5)^2 - (n - 5)^2 = 20n$ for every value of n

[2 marks]

F2. A rectangle is $(x + 6)$ cm long and $(x - 2)$ cm wide. Its area is 33 cm^2.
(a) Show that $x^2 + 4x - 45 = 0$. (b) Solve the equation to find the width of the rectangle.

[4 marks]

F3. Without a calculator, use the difference of two squares to work out $53^2 - 47^2$

[1 mark]